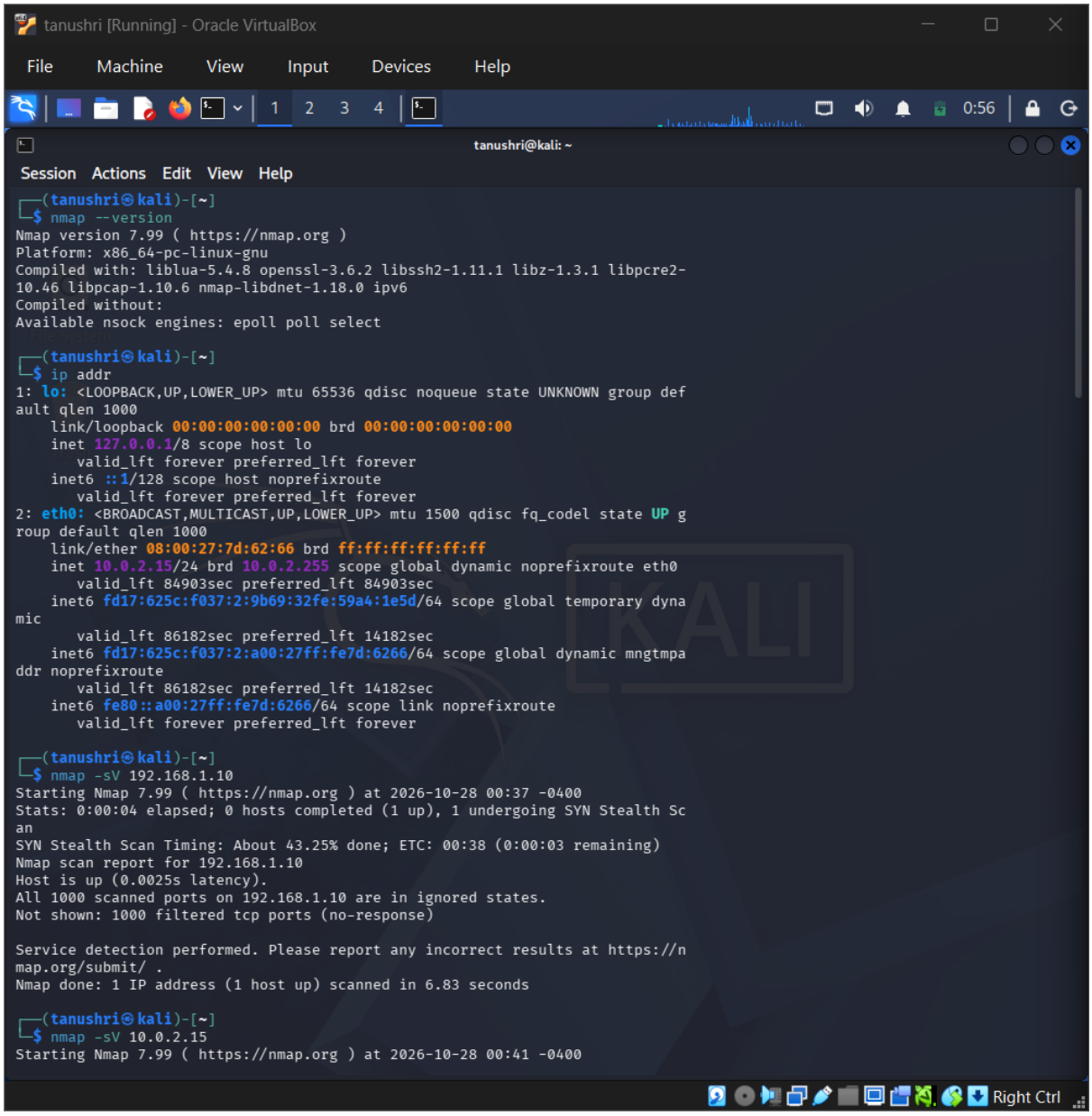
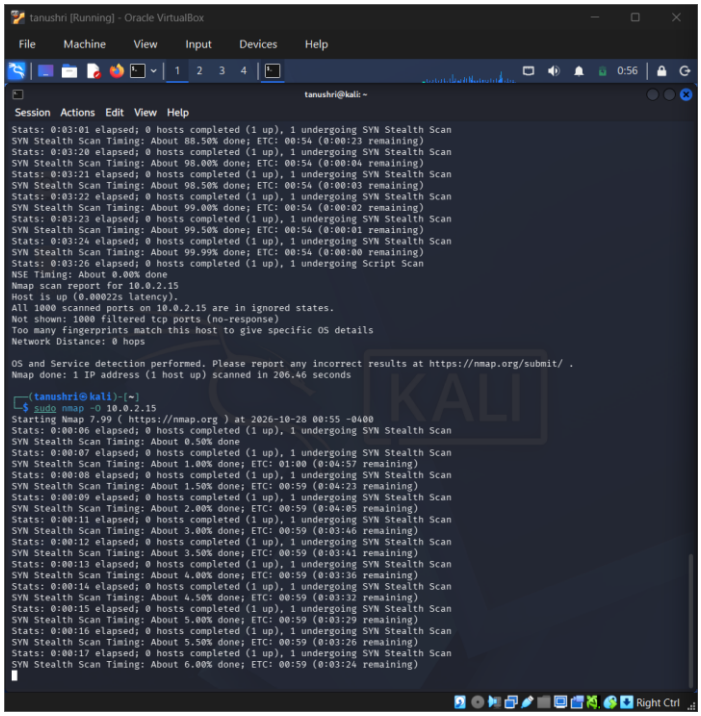
LAB MANUAL
Ethical Hacking Laboratory – Kali Linux Commands and Nmap Scanning
Course Code: Ethical Hacking Laboratory
Software Required: Kali Linux (Latest Version)

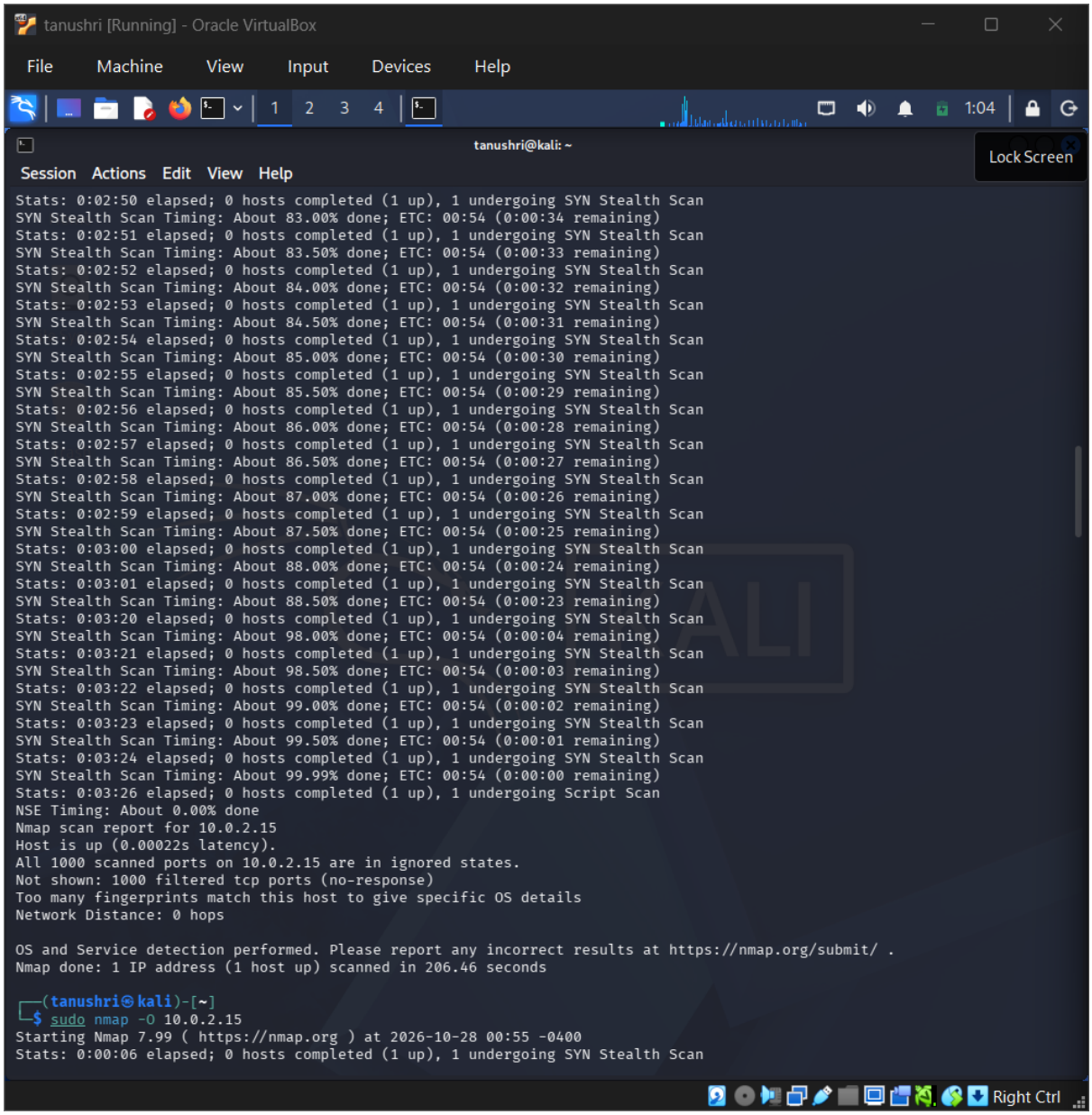
Experiment 8

Nmap Service Version and OS Detection
Aim
To perform service version detection and operating system identification using Nmap.
Theory
Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.
Commands
i) Detect Service Versions
nmap -sV 192.168.1.10
Description:
Determines the versions of services running on target ports.
ii) Aggressive Scan
nmap -A 192.168.1.10
Description:
Performs OS detection, version detection, script scanning, and traceroute.
iii) Operating System Detection
sudo nmap -O 192.168.1.10
Description:
Identifies the operating system running on the target host.
Sample Output
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 8.2
80/tcp open http Apache 2.4
Result
Successfully performed service version detection and OS identification using Nmap.

Output:


tanushri [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
0:56
tanushri@kali: ~
Session Actions Edit View Help
(tanushri㉿kali)-[~]
$ nmap --version
Nmap version 7.99 (https://nmap.org)
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
(tanushri㉿kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:7d:62:66 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
valid_lft 84903sec preferred_lft 84903sec
inet6 fd17:625c:f037:2:9b69:32fe:59a4:1e5d/64 scope global temporary dynamic
valid_lft 86182sec preferred_lft 14182sec
inet6 fd17:625c:f037:2:a00:27ff:fe7d:6266/64 scope global dynamic mngtmpaddr noprefixroute
valid_lft 86182sec preferred_lft 14182sec
inet6 fe80::a00:27ff:fe7d:6266/64 scope link noprefixroute
valid_lft forever preferred_lft forever
(tanushri㉿kali)-[~]
$ nmap -sV 192.168.1.10
Starting Nmap 7.99 (https://nmap.org) at 2026-10-28 00:37 -0400
Stats: 0:00:04 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 43.25% done; ETC: 00:38 (0:00:03 remaining)
Nmap scan report for 192.168.1.10
Host is up (0.0025s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.83 seconds
(tanushri㉿kali)-[~]
$ nmap -sV 10.0.2.15
Starting Nmap 7.99 (https://nmap.org) at 2026-10-28 00:41 -0400
Right Ctrl


tanushri [Running] - Oracle VirtualBox
File Machine View Input Devices Help
tanushri@kali: ~
Session Actions Edit View Help
Stats: 0:03:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 88.50% done; ETC: 00:54 (0:00:23 remaining)
Stats: 0:03:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 98.00% done; ETC: 00:54 (0:00:04 remaining)
Stats: 0:03:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 98.50% done; ETC: 00:54 (0:00:03 remaining)
Stats: 0:03:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.00% done; ETC: 00:54 (0:00:02 remaining)
Stats: 0:03:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.50% done; ETC: 00:54 (0:00:01 remaining)
Stats: 0:03:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:54 (0:00:00 remaining)
Stats: 0:03:26 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 0.00% done
Nmap scan report for 10.0.2.15
Host is up (0.00022s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 206.46 seconds
(tanushri@kali)-[~]
$ sudo nmap -O 10.0.2.15
Starting Nmap 7.99 (https://nmap.org) at 2026-10-28 00:55 -0400
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.50% done
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.00% done; ETC: 01:00 (0:04:57 remaining)
Stats: 0:00:08 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.50% done; ETC: 00:59 (0:04:23 remaining)
Stats: 0:00:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 2.00% done; ETC: 00:59 (0:04:05 remaining)
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.00% done; ETC: 00:59 (0:03:46 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.50% done; ETC: 00:59 (0:03:41 remaining)
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 4.00% done; ETC: 00:59 (0:03:36 remaining)
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 4.50% done; ETC: 00:59 (0:03:32 remaining)
Stats: 0:00:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.00% done; ETC: 00:59 (0:03:29 remaining)
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 5.50% done; ETC: 00:59 (0:03:26 remaining)
Stats: 0:00:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.00% done; ETC: 00:59 (0:03:24 remaining)
Right Ctrl


tanushri [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
1:04
tanushri@kali: ~
Lock Screen
Session Actions Edit View Help
Stats: 0:02:50 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 83.00% done; ETC: 00:54 (0:00:34 remaining)
Stats: 0:02:51 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 83.50% done; ETC: 00:54 (0:00:33 remaining)
Stats: 0:02:52 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.00% done; ETC: 00:54 (0:00:32 remaining)
Stats: 0:02:53 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.50% done; ETC: 00:54 (0:00:31 remaining)
Stats: 0:02:54 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.00% done; ETC: 00:54 (0:00:30 remaining)
Stats: 0:02:55 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.50% done; ETC: 00:54 (0:00:29 remaining)
Stats: 0:02:56 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 86.00% done; ETC: 00:54 (0:00:28 remaining)
Stats: 0:02:57 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 86.50% done; ETC: 00:54 (0:00:27 remaining)
Stats: 0:02:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 87.00% done; ETC: 00:54 (0:00:26 remaining)
Stats: 0:02:59 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 87.50% done; ETC: 00:54 (0:00:25 remaining)
Stats: 0:03:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 88.00% done; ETC: 00:54 (0:00:24 remaining)
Stats: 0:03:01 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 88.50% done; ETC: 00:54 (0:00:23 remaining)
Stats: 0:03:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 98.00% done; ETC: 00:54 (0:00:04 remaining)
Stats: 0:03:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 98.50% done; ETC: 00:54 (0:00:03 remaining)
Stats: 0:03:22 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.00% done; ETC: 00:54 (0:00:02 remaining)
Stats: 0:03:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.50% done; ETC: 00:54 (0:00:01 remaining)
Stats: 0:03:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.99% done; ETC: 00:54 (0:00:00 remaining)
Stats: 0:03:26 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 0.00% done
Nmap scan report for 10.0.2.15
Host is up (0.00022s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 206.46 seconds
┌──(tanushri㉿kali)-[~]
└─$ sudo nmap -O 10.0.2.15
Starting Nmap 7.99 (https://nmap.org) at 2026-10-28 00:55 -0400
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
Right Ctrl